

DATA SOURCES

SemEval ASTE: Rest14 (train), Rest14/15/16/Laptop14 (test)

DMASTE: 4 source domains (electronics, beauty, fashion, home) → 4 target (book, grocery, pet, toy)

CANONICAL FORMAT

{sentence, tokens, annotations[{aspect, opinion, polarity}]}

AUGMENTATION (optional)

Aspect masking (configurable fraction, replace with ;extra_id;)

Duplication (controlled data multiplication)

Cross-domain data mixing (DMASTE fractions added to Rest14)

SYNTAX ENRICHMENT (optional)

Compact dependency (separate Syntax: line, content words only)

Compact POS (POS tags for content words)

Inline (dep. or POS annotations mixed into input tokens)

GENERATIVE FORMAT CONVERSION

Task subset: full triplet — singles — pairs

Curriculum: waypoint interpolation across epochs

Output format:

Structured: [aspect, opinion, polarity]

NL: “{aspect} is described as {opinion}, expressing a {polarity} sentiment”

MvP markers: [A] x [O] y [S] z [SSEP]

Multi-ordering: 6 permutations of element order

STAR: pairwise sub-task decomposition ($7 \times$ data)

Per-epoch re-randomisation (all above re-executed each epoch with different seed)

FLAN-T5-base (250M)

AdamW, lr=3e-4, cosine, 30 epochs

Label smoothing=0.05, greedy

RoBERTa BASELINE (125M)

BIO tagger + biaffine scorer

+ polarity classifier

OUTPUT PARSING

NL: regex template matching

Structured: bracket/comma splitting

MvP: [SSEP] separator + marker extraction

Malformed outputs → score 0 (strict, no partial credit)

EVALUATION

Exact-match micro P/R/F1 (primary metric)

Macro P/R/F1

Lenient F1 (token overlap, threshold=0.8)

Lenient F1 (longest common subsequence, threshold=0.8)

Soft F1 (embedding cosine similarity, aspect key, all-MiniLM-L6-v2)

Average token overlap + LCS overlap + match rate

Multi-seed protocol: 5 seeds (42, 123, 456, 789, 1337), mean $\pm$ std

DATA SOURCES

English: SemEval ASTE — Rest14 (train), Rest14/15/16/Laptop14 (test)

Romanian: eMAG phone reviews — 27,630 train / 3,093 test, GPT-4 annotated, 11 categories

CANONICAL FORMAT: {sentence, tokens, annotations[{aspect, opinion, polarity, category}]}

GENERATIVE PATH

NL format conversion
(Romanian template)

FLAN-T5-base (250M)
lr=3e-4, cosine, 12 epochs

Greedy decoding

CLASSIFIER PATH

Tokenise + Encode:

RoBERTa-base (114M, 12 layers)

bert-ro-cased (110M, RoBERTa)

XLM-R-base (278M, multilingual)

Classification heads:

Polarity (linear)

LLM INFERENCE PATH

Demo retrieval:

3 × BM25 + 3 × SimCSE

Prompt construction:

0-shot (task + taxonomy)

6-shot (+ 6 examples)

Generation (41% success rate):

OUTPUT PARSING

Generative: regex template matching (NL format)

Classifier: argmax per head (polarity, category)

LLM: regex parsing, same strict protocol (malformed → score 0)

EVALUATION

English ASTE: Exact-match triplet micro-F1 — Lenient F1 (token overlap + LCS)

Soft F1 (embedding cosine) — Avg. overlap + match rate

Romanian: Polarity + Category pair F1 — Polarity F1 — Category F1

Cross-method comparison: fine-tuned (110–278M) vs LLM (27–35B)